

Processing Speed -- Guided Notes ANSWER KEY

Unit tp-1.7 | CompTIA Network+ N10-009 Obj. FC0-U71 1.3 | N10-008 FC0-U71 1.3

For instructor use / distribute after students complete the notes.

Question 1

Hz stands for _____ and represents one cycle per second. GHz stands for _____ and equals _____ cycles per second. A CPU running at 3.6 GHz completes approximately _____ billion instruction cycles per second.

Answer: Hertz; gigahertz; 1,000,000,000 (one billion); 3.6 billion.

Question 2

Clock speed _____ (always/does not always) determine overall CPU performance. A CPU running single-threaded workloads benefits from _____ clock speed, while tasks that can be divided into parallel work benefit from more _____ even at lower clock speeds.

Answer: Does not always; higher; cores.

Question 3

A multi-core CPU has multiple _____ on a single chip, each able to execute instructions _____ (sequentially/independently). A 4-core CPU can technically work on _____ different tasks at the same time, improving _____ when running multiple programs simultaneously.

Answer: Cores; independently; 4; multitasking (throughput).

Question 4

CPU cache is ultra-fast memory built directly into the processor. _____ cache is the smallest and fastest, closest to the core. _____ cache is shared within a core and slightly larger. _____ cache is the largest and shared among all cores. Cache reduces how often the CPU must wait for data from _____, which is much slower.

Answer: L1; L2; L3; RAM (main memory).

Question 5

Thermal throttling occurs when a CPU reaches its maximum safe _____ and automatically reduces its _____ to generate less heat. While this prevents hardware damage, it results in _____ performance. The long-term fix is to improve _____ or reapply thermal paste.

Answer: Temperature; clock speed; reduced (degraded); cooling.

Question 6

A _____ is a standardized test used to measure and compare CPU, GPU, or system performance. The score allows comparisons between different hardware. However, benchmark results _____ (always/do not always) reflect real-world performance for every type of workload.

Answer: Benchmark; do not always.

Question 7

Hyper-Threading (Intel) and SMT (AMD) allow each physical CPU core to handle _____ threads simultaneously. A 4-core CPU with hyper-threading appears to the OS as _____ logical processors. This improves performance when the CPU has _____ work waiting than physical cores to run it.

Answer: Two; 8 logical processors; more.

Question 8 -- Real World Application

REAL WORLD: A student is choosing between two laptops. Laptop A has a 4-core CPU at 4.0 GHz. Laptop B has an 8-core CPU at 2.8 GHz. The student assumes Laptop A is faster because it has a higher clock speed. For which tasks would Laptop A actually perform better, and for which would Laptop B perform better? Explain your reasoning.

Answer: Laptop A (4-core, 4.0 GHz) performs better on single-threaded tasks -- those that cannot be parallelized -- such as older games that rely on one CPU core, some legacy applications, and tasks where instructions must run sequentially. The higher clock speed means each individual core completes cycles faster. Laptop B (8-core, 2.8 GHz) performs better on multi-threaded workloads such as video rendering, compiling code, running multiple virtual machines, streaming while gaming, and modern games optimized for many cores. The lower clock speed is offset by having twice as many cores working in parallel. A higher GHz number alone does not make a CPU faster for all tasks.